

Solution Requirements

Date	3 July 2026
Team ID	XXXXXX
Project Name	Rising Waters-Flood Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The application allows users to securely access the flood prediction system (optional login if implemented; otherwise authorized local access).</i>	Medium
2	Authorization levels	<i>Users can enter weather parameters and view predictions. Administrators can update the trained model and maintain the system.</i>	Medium

3	External interfaces	<i>The system uses a historical weather dataset, Flask framework, and machine learning libraries (Scikit-learn, Pandas, NumPy).</i>	
4	Transactions processing	<i>The system accepts weather inputs, processes them using the trained Decision Tree model, and returns a flood prediction instantly.</i>	High
5	Reporting	<i>The application displays prediction results and model accuracy. Prediction history may be stored for analysis.</i>	Medium
6	Business rules, etc.	<i>The system predicts flood risk only after validating all required weather inputs. Invalid or missing values are rejected.</i>	High
7	Compliance to laws or regulations	<i>The application should protect user information and follow basic data privacy and ethical AI practices.</i>	Medium
8	Other	<i>The trained Decision Tree model is saved and loaded automatically during prediction.</i>	High

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	<i>The system should generate predictions quickly.</i>	Prediction displayed in less than 2 seconds
2	Scalability	<i>The application should support multiple prediction requests with minimal delay.</i>	Supports 100+ prediction requests without performance issues
3	Security & Data Privacy	<i>Input data should be handled securely and protected from unauthorized access.</i>	Basic input validation and secure application deployment
4	Reliability & Availability	<i>The application should provide consistent and accurate predictions whenever it is running.</i>	99% application availability
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy to use on desktop and mobile devices.</i>	Users can complete a prediction in 3 simple steps

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

6	<i>Other</i>	<i>Maintainability & Portability</i>	Code is modular, documented, and can run on Windows, Linux, macOS, and IBM Cloud.
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